Practice problems #16

1. Let $y = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$, $u_1 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$, $u_2 = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$, and $W = \text{Span}\{u_1, u_2\}$.
   Let $U = [u_1\ u_2]$. Given that the matrix U has orthonormal columns, compute $UU^T y$ *without* carrying out the matrix multiplication directly.

2. Consider the matrix
$$A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$
   (a) Show that A has linearly independent columns.
   (b) Give an orthogonal and orthonormal bases for the column space of A.

Answers or Hints on page 2.

#1: Note that it is an orthogonal projection of something onto something. Ans: $\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$.

#2(a): Show!

#2(b): Through Gram-Schmidt, you get $u_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$,

and

$$\left\{ \frac{u_1}{||u_1||}, \quad \frac{u_2}{||u_2||}, \quad \frac{u_3}{||u_3||} \right\},$$

where $||u_1|| = ||u_3|| = 2$, and $||u_2|| = 2\sqrt{2}$.